

Examples of Interactive Use of PAARL

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We illustrate the interactive use of PAARL on the Traffic Norms Domain, exemplified in Figure 1, in which a self-driving agent navigates a grid-layout city while pursuing either emergency or non-emergency goals. The environment contains traffic signs (e.g., Do Not Enter and Stop signs), speed limits, traffic lights, pedestrians, and stopped school buses. The agent can execute actions of the form $drive(L_1, L_2, S)$, representing movement between connected locations L_1 and L_2 at speed $S > 0$, and $stop(L)$, representing stopping at location L .

The environment is governed by a collection of traffic policies. These include speed-limit regulations with associated penalties, obligations to stop for pedestrians and school buses (with high penalties for non-compliance), and policies governing traffic signs and traffic lights.

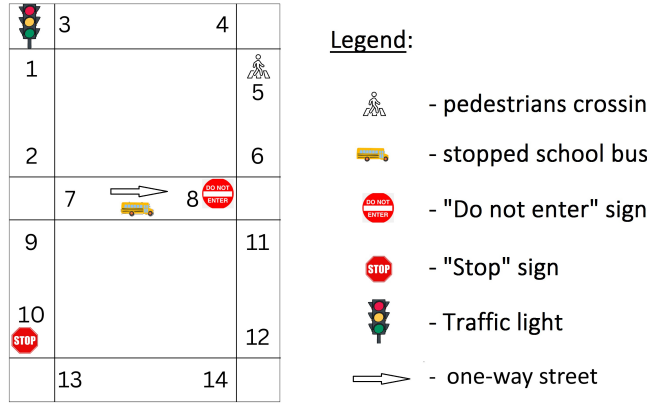


Fig. 1. Layout of the Traffic Norms Domain

1 Example 1

To define a scenario in which the agent starts at location 5 and must reach location 10 in an emergency situation, the PAARL interface is initialized as shown in Figure 2. The user specifies the start and goal locations, execution mode,

and any additional observations in the left panel. Pressing the *Start New Scenario* button generates the planning instance displayed on the top-right side of the interface together with the corresponding `clingo` output shown below it. The resulting answer set contains the planned agent actions represented by *agent_occurs/2* atoms, together with incurred penalties, cumulative completion time, and cumulative penalty. Since the scenario prioritizes minimizing completion time over policy compliance, the generated plan violates several traffic policies. For example, penalties are incurred for exceeding the speed limit (policy *r1*), as indicated by atoms such as *add_penalty(r1(5, 6, 85, 25), 3, 0)*, representing a 3-point penalty incurred by driving at time 0 from location 5 to 6 at 85 mph when the speed limit is 25 mph. Another penalty is incurred at time step 1 for violating policy *r3* governing Do Not Enter signs when driving from location 6 to 8: *add_penalty(r3(6, 8, 45), 3, 1)*.

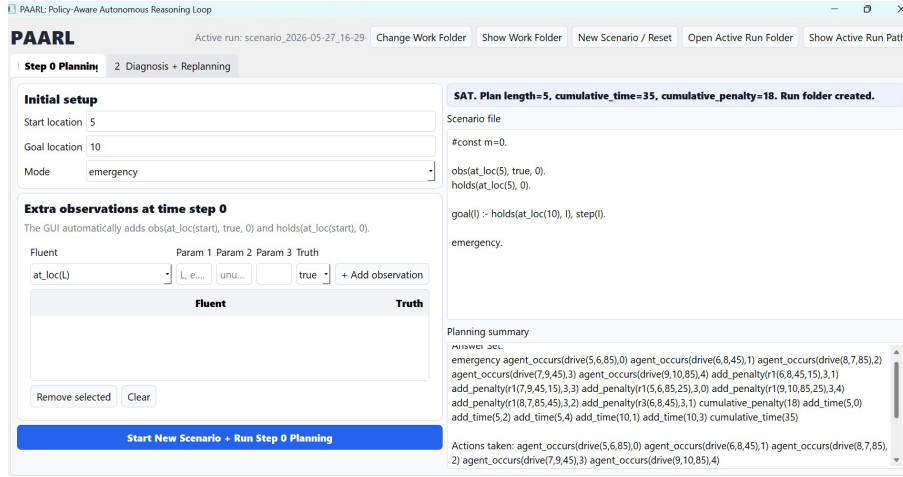


Fig. 2. PAARL System: Initial Time Step for Example 1

Figure 3 illustrates entering new observations at time step 2. Upon reaching location 8, the agent observes a stopped school bus between locations 8 and 7. According to the traffic policies, failing to stop in this situation incurs a very high penalty as not stopping may directly endanger humans; thus compliance is expected even in emergency scenarios. To record the new observations, the PAARL interface opens a *New observation step* panel where the user enters the current time step ($m = 2$), and an *Observations at time step m* panel where the user enters observations through a drop-down menu populated with available fluents. Since the agent did not observe the school bus stopping, but only observed it already stopped, pressing the *New Diagnosis + Planning for Time Step m* button triggers both diagnosis and re-planning. PAARL generates the diagnosis file and the updated planning instance. The corresponding outputs appear in the

Diagnosis summary and *Planning summary* sections of the interface. Diagnosis explains the observation by hypothesizing that the school bus stopped earlier, at time step 1, while the updated plan instructs the agent to stop at location 8 at time step 2: *agent_occurs(stop(8), 2)*. Further observations can be entered after this step, until the agent reaches its goal. In the top-right corner of the PAARL window, buttons allow users to manage scenario log folders and start new scenarios.

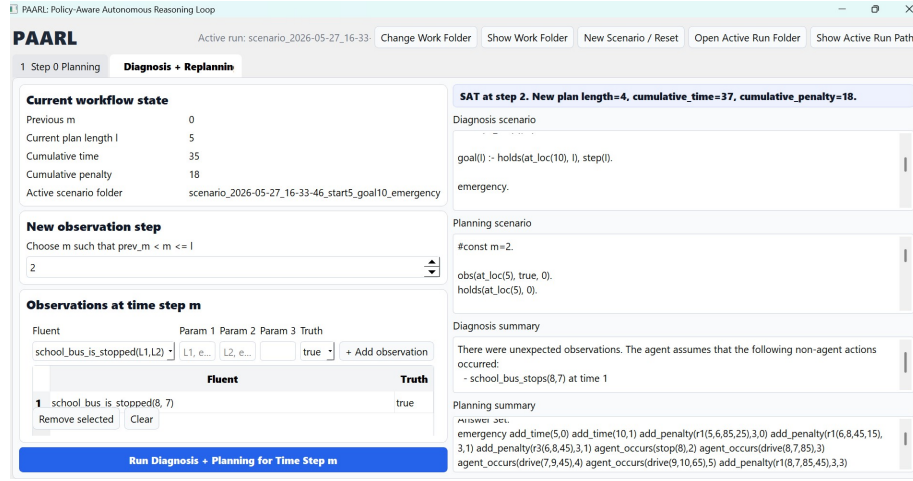


Fig. 3. PAARL System: New Observations and Re-planning at Time Step 2

2 Example 2

To define a second scenario in which the agent starts at location 6 and must reach location 1 in an emergency situation, the PAARL interface is initialized as shown in Figure 4. The user specifies the start and goal locations, selects the emergency execution mode, and presses the **Start New Scenario + Run Step 0 Planning** button. PAARL then generates the planning instance displayed in the upper-right portion of the interface, together with the corresponding *clingo* output shown in the planning summary below it. The resulting answer set contains the planned agent actions represented by *agent_occurs/2* atoms, together with incurred penalties, cumulative completion time, and cumulative penalty.

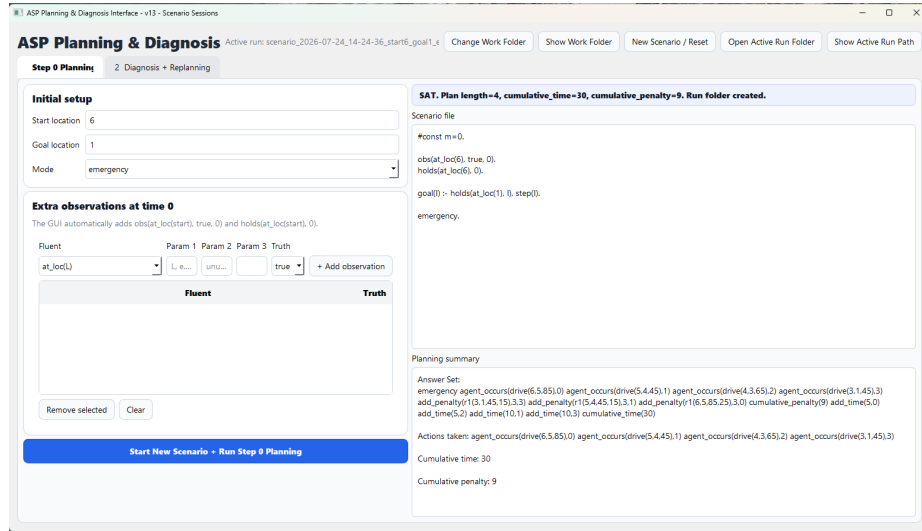


Fig. 4. PAARL System: Initial Time Step for Example 2.

The initial plan directs the agent along the route $6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 1$ using the following sequence of actions:

$$\begin{aligned}
 &agent_occurs(drive(6, 5, 85), 0), \\
 &agent_occurs(drive(5, 4, 45), 1), \\
 &agent_occurs(drive(4, 3, 65), 2), \\
 &agent_occurs(drive(3, 1, 45), 3).
 \end{aligned}$$

Since the scenario prioritizes minimizing completion time over policy compliance, the generated plan violates several speed-limit policies. At time step 0, the agent drives from location 6 to location 5 at 85 mph, although the speed limit on this road segment is 25 mph. The resulting violation is represented by the atom

$$add_penalty(r1(6, 5, 85, 25), 3, 0).$$

Another penalty is incurred at time step 1 when the agent travels from location 5 to location 4 at 45 mph in a 15 mph zone:

$$add_penalty(r1(5, 4, 45, 15), 3, 1).$$

A third penalty is incurred at time step 3 when the agent drives from location 3 to location 1 at 45 mph, although the speed limit on that road segment is 15 mph:

$$add_penalty(r1(3, 1, 45, 15), 3, 3).$$

Each speed-limit violation incurs a penalty of 3, resulting in a cumulative penalty of 9. The initial plan reaches the goal in a cumulative time of 30.

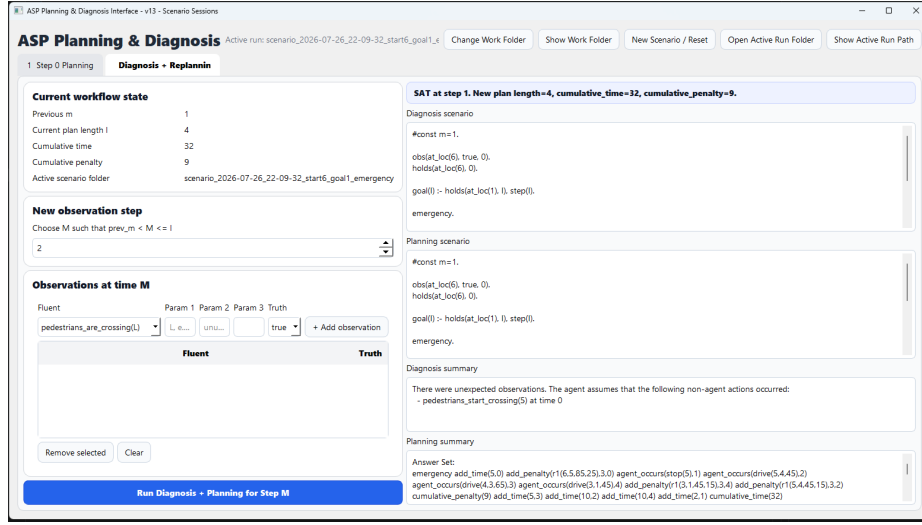


Fig. 5. PAARL System: Diagnosis and Re-planning at Time Step 1 for Example 2.

Figure 5 illustrates the introduction of a new observation at time step $m = 1$. After executing the first action and reaching location 5, the agent observes that pedestrians are crossing at that location. To record this information, the user selects time step $m = 1$ in the **New observation step** panel and enters *pedestrians_are_crossing(5)* in the **Observations at time M** panel.

Since the agent did not observe the pedestrians beginning to cross, but only observed that they were already crossing, the new observation is unexpected with respect to the previously recorded history. Pressing the **Run Diagnosis + Planning for Step M** button therefore triggers both diagnosis and re-planning.

The diagnosis summary shown in Figure 5 explains the unexpected observation by hypothesizing that the following non-agent action occurred:

$$pedestrians_start_crossing(5)$$

at time step 0. This hypothesized action explains why the fluent *pedestrians_are_crossing(5)* holds at time step 1, even though the agent did not directly observe when the pedestrians began crossing.

After incorporating the diagnosis and the action already executed at time step 0, PAARL generates an updated plan from the agent's current state. The revised plan first instructs the agent to stop at location 5 at time step 1:

$$agent_occurs(stop(5), 1).$$

The agent subsequently continues toward the goal by executing

$$\begin{aligned} &agent_occurs(drive(5, 4, 45), 2), \\ &agent_occurs(drive(4, 3, 65), 3), \\ &agent_occurs(drive(3, 1, 45), 4). \end{aligned}$$

Thus, in contrast to the initial plan, the revised plan inserts a stop action at location 5 so that the agent complies with the policy governing pedestrian crossings. The stop requires two additional time units, as represented by

$$add_time(2, 1),$$

and consequently increases the cumulative completion time from 30 to 32. The cumulative penalty remains 9 because the revised plan retains the three speed-limit violations associated with the selected route while avoiding an additional violation of the pedestrian-crossing policy.

This example demonstrates PAARL's ability to incorporate an unexpected observation during plan execution, diagnose an unobserved exogenous event that explains the observation, preserve the already executed portion of the trajectory, and revise the remaining plan so that the agent responds appropriately to a safety-critical situation.